

WYNYARD, SK, Canada

WMO: 718650

Lat: 51.770N

Lon: 104.200W

Elev: 1838

StdP: 13.75

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-21.7	-34.1	0.9	-25.9	-28.8	1.2	-21.1	24.7	16.1	22.6	16.4	6.5	260	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	83.4	65.6	80.1	64.1	77.1	62.3	68.7	79.4	66.4	76.8	64.4	73.8	9.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	98.3	74.8	62.4	90.4	71.8	60.5	84.1	69.7	34.0	79.1	32.1	76.8	30.5	73.9	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.3	19.9	17.8	DB	-33.1	90.2	5.6	3.9	-37.1	93.0	-40.4	95.3	-43.5	97.5	-47.6	100.3
			WB	-33.1	72.8	5.3	2.7	-37.0	74.8	-40.1	76.4	-43.1	78.0	-46.9	79.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.9	4.9	7.6	19.6	37.0	50.1	59.3	64.1	62.1	53.5	39.1	22.4	9.7
	DBStd	24.09	16.07	14.63	14.30	10.83	8.94	6.58	5.69	6.27	8.48	9.43	12.63	14.74
	HDD50	6544	1399	1187	943	405	111	7	0	1	58	356	828	1249
	HDD65	10742	1864	1607	1408	840	465	191	86	132	355	803	1277	1714
	CDD50	1411	0	0	0	15	115	287	437	375	162	19	1	0
	CDD65	134	0	0	0	0	5	21	58	41	9	0	0	0
	CDH74	1400	0	0	0	4	74	218	537	428	137	2	0	0
	CDH80	298	0	0	0	1	12	40	117	102	26	0	0	0
Wind	WSAvg	8.4	8.6	8.5	8.9	9.2	9.1	8.0	7.3	7.2	8.2	8.7	8.9	8.6
Precipitation	PrecAvg	17.7	0.8	0.5	0.6	1.1	1.7	3.3	2.9	2.6	1.7	1.2	0.8	0.7
	PrecMax	27.2	1.6	1.1	1.5	2.9	4.1	7.4	5.7	6.9	3.9	3.2	1.8	1.7
	PrecMin	10.1	0.2	0.1	0.1	0.2	0.1	0.7	1.3	0.3	0.1	0.0	0.1	0.2
	PrecStd	4.1	0.4	0.2	0.4	0.8	1.0	1.5	1.1	1.6	0.9	0.8	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.9	39.5	52.2	69.9	81.2	84.9	87.5	88.4	83.9	71.3	55.5	41.8
		MCWB	35.6	35.1	42.8	52.4	58.2	64.9	70.4	66.8	61.8	55.3	45.4	36.5
	2%	DB	36.0	35.0	45.5	64.4	75.3	79.8	83.2	82.3	78.0	63.5	47.6	37.6
		MCWB	32.7	32.1	39.2	49.0	55.4	62.9	67.8	64.4	59.6	50.5	40.8	33.3
	5%	DB	32.0	31.8	40.5	59.1	70.9	76.0	79.8	78.7	72.9	57.9	42.2	33.4
		MCWB	29.5	29.6	36.0	46.8	53.6	61.0	66.2	63.6	57.6	47.7	37.0	30.8
	10%	DB	27.5	28.3	37.0	53.6	66.7	72.3	76.6	75.1	68.1	52.9	37.8	29.5
		MCWB	25.7	26.5	33.7	43.6	51.8	58.6	64.2	61.7	55.5	45.3	34.1	27.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.8	35.5	43.4	53.6	60.8	69.8	72.8	69.7	64.6	55.8	45.7	36.8
		MCDB	39.5	38.3	50.4	67.6	74.3	82.0	84.4	81.5	77.1	68.4	54.2	41.1
	2%	WB	32.8	32.4	39.3	50.4	57.6	65.2	70.0	67.1	61.4	51.8	41.0	33.5
		MCDB	35.7	34.7	44.9	62.2	71.4	75.3	80.3	78.7	74.0	60.2	46.9	37.0
	5%	WB	29.5	29.6	36.3	47.3	55.3	62.8	67.6	65.3	59.1	48.6	37.5	30.8
		MCDB	31.9	31.8	40.4	57.3	68.8	72.7	77.7	76.1	70.7	56.5	41.6	33.6
	10%	WB	25.8	26.5	33.9	44.3	52.9	60.6	65.4	63.2	56.4	45.8	34.3	27.5
		MCDB	27.5	28.3	36.7	52.9	64.1	69.8	74.5	72.6	66.6	52.3	37.6	29.5
Mean Daily Temperature Range	5% DB	MDBR	15.7	16.0	16.0	18.5	21.7	18.7	19.6	20.5	20.0	16.4	13.7	14.4
		MCDBR	17.0	16.4	17.8	26.8	27.2	24.6	24.8	25.4	27.5	24.5	18.5	16.5
		MCWBR	14.5	14.3	13.0	15.2	12.9	11.8	12.9	12.3	13.3	14.2	12.7	13.8
	5% WB	MCDBR	16.6	15.2	17.3	25.0	24.0	20.5	22.6	22.7	24.9	22.8	17.6	16.5
		MCWBR	14.6	13.3	12.7	14.9	12.2	11.3	12.9	12.1	12.9	13.8	12.6	14.0
Clear-Sky Solar Irradiance	taub		0.242	0.288	0.287	0.321	0.328	0.348	0.358	0.362	0.313	0.279	0.249	0.255
	taud		2.311	2.173	2.356	2.419	2.450	2.429	2.398	2.384	2.512	2.558	2.477	2.246
	Ebn at Noon		251	262	291	292	293	287	282	274	279	271	252	219
	Edh at Noon		19	30	30	33	33	35	35	34	26	21	16	17
All-Sky Solar Radiation	RadAvg		377	718	1190	1542	1857	1890	1976	1605	1173	682	343	266
	RadStd		33	50	99	125	158	140	112	121	101	95	46	28

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page